Examiner only

9. (a) Addition of a solution containing $Ag^+(aq)$ to solutions containing halide ions such as chloride, bromide or iodide causes precipitates to form.

(i) Give the colours of the precipitates formed with chloride, bromide and iodide ions. [1]

chloride ions ....

bromide ions ....

iodide ions ....

(ii) The standard enthalpy change of formation of AgCl(s) is –127.1 kJ mol^{-1}.

Use the data below to calculate the standard enthalpy change of formation of AgBr(s) and hence show which of the two silver halides is more stable with respect to its elements. You **must** show your working. [4]

Reaction	Standard enthalpy change, ΔH^θ / kJ mol^{-1}
$Ag(s) \longrightarrow Ag^+(aq) + e^-$	+105.6
$Br_2(l) + 2e^- \longrightarrow 2Br^-(aq)$	–243.1
$AgBr(s) \longrightarrow Ag^+(aq) + Br^-(aq)$	+84.4

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(b) The standard electrode potential for the half-equation below is +0.80 V.

$$Ag^{+}(aq) \quad + \quad e^{-} \quad \rightleftharpoons \quad Ag(s)$$

(i) When a piece of copper metal is placed in a solution of silver nitrate, a displacement reaction occurs.

I. Write the **ionic** equation for this displacement reaction. [1]

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II. State what information this provides regarding the standard electrode potential for the $Cu(s)|Cu^{2+}(aq)$ half-cell. Give a reason for your answer. [2]

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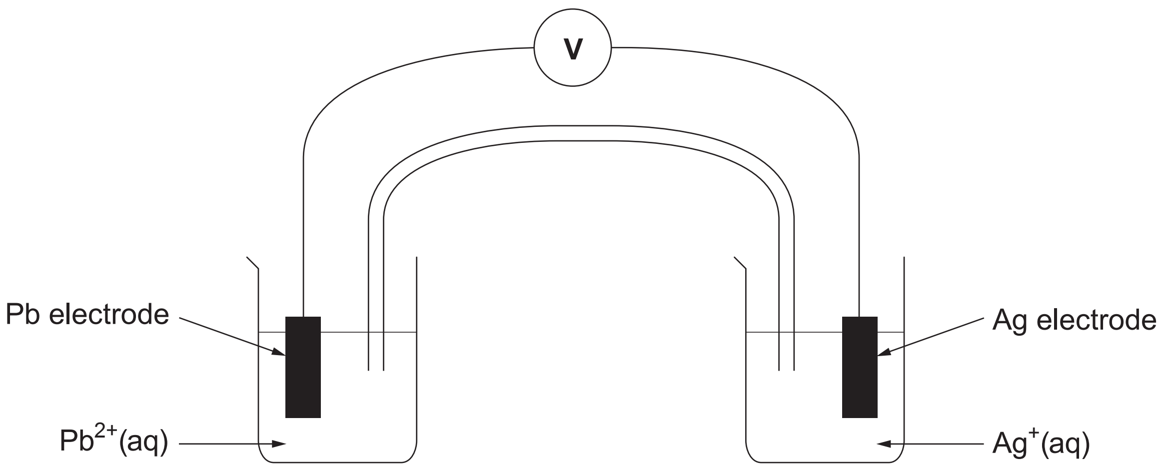
(ii) The apparatus below was assembled with the $Pb(s)|Pb^{2+}(aq)$ half-cell under standard conditions connected to the $Ag(s)|Ag^{+}(aq)$ half-cell. The silver is the positive electrode.



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The value on the high-resistance voltmeter was recorded with different concentrations of silver ions used in the Ag(s)|Ag^+(aq) half-cell. The results are shown below.

Concentration of Ag^+(aq) / $mol\,dm^{-3}$	Value recorded on high-resistance voltmeter / V
1.0	0.93
0.1	0.87
0.01	0.81
0.001	0.75
0.0001	0.69

I. Calculate the value of the standard electrode potential for the Pb(s)|Pb^{2+}(aq) half-cell. [2]

E^θ = ... V

II. Explain why the value recorded on the voltmeter becomes less positive as the concentration of silver ions decreases. [3]

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